

① $\dot{x} = u - u^2$, $|u| \leq 1$, $\int_0^1 |x(t)| dt$, $x(1) = -1 \rightarrow x(t_f) = 0$.
 $J = \int_0^1 |x(t)| dt$
 $u = ?$ $x = ?$ $t_f = ?$

$x_1(t_f) = 0$ $x_1(0) = -1$
 $x_2(t_f) = 0$ $x_2(0) = -1$

점수:
 914 602 0259
 9we1234

$\dot{x}_1 = x_2$
 $\dot{x}_2 = -2x_1$

$\exists$ 1) state x_0 on M 나중 state 변경되지 않는다. $\Rightarrow$ 최적시간.
 $x = \int x_2 = u - u^2 = x$

$H = \mathcal{L} + \lambda f = t + \lambda_1 \dot{x}_1 + \lambda_2 \dot{x}_2 = t + \lambda_1 x_2 + \lambda_2 (-2x_1)$

$\left[\frac{\partial H}{\partial x} + D^T \frac{\partial W}{\partial x} - \lambda^T \right]_{t=t_f} \delta x + \left[H + \frac{\partial H}{\partial t} + D^T \frac{\partial W}{\partial t} \right]_{t=t_f} \delta t = 0 \rightarrow H(t_f) = 0, \dot{H} = 0 \rightarrow H(t) = \text{const} = H(t_f) = 0, \therefore H(t) = 0$.

$\dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = 0 \rightarrow \lambda_1 = \text{const}$. If $\lambda_2 = \oplus$ $u = -u_0$ $u = -u_0 \text{sign}(\lambda_2)$
 $\dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -\lambda_1 \rightarrow \lambda_2 = -\lambda_1 t + \lambda_{20}$ $\lambda_2 = \ominus$ $u = +u_0$

If $u_0 = 1$, $\dot{x}_1 = x_2$ $\dot{x}_2 = -1$ or $\dot{x}_1 = x_2$ $\dot{x}_2 = 3$ $\rightarrow \frac{dx_1}{dx_2} = -x_2, 3x_2$

$\int dx_1 = \int -x_2 dx_2, \int dx_1 = \int 3x_2 dx_2 = \frac{1}{2} x_2^2 + C_2$

$x_1 = -\frac{1}{2} x_2^2 + C_1$

$x_1 = -\frac{1}{2} x_2^2 + C_1$ ($u=1$)

$x_1 = \frac{3}{2} x_2^2 + C_2$ ($u=-1$)

If $\begin{cases} x_2 = \oplus, x_1 = \frac{3}{2} x_2^2 + C_2 \\ x_2 = \ominus, x_1 = -\frac{1}{2} x_2^2 + C_1 \end{cases} \rightarrow x_1(0) = \frac{3}{2} x_2(0)^2 + C_2 \rightarrow -1 = \frac{3}{2} (-1)^2 + C_2 \rightarrow C_2 = -0.5$
 $x_1(0) = -\frac{1}{2} x_2(0)^2 + C_1 \rightarrow -1 = -\frac{1}{2} (-1)^2 + C_1 \rightarrow C_1 = -0.5$

$x_1 = \frac{3}{2} x_2^2 - 0.5$
 $x_1 = -\frac{1}{2} x_2^2 - 0.5$

$S = \frac{1}{2} x_2^2 + 1 |x_2| x_2 - 0.5$

$\therefore u(t) = \begin{cases} 1 & S = \oplus \\ -1 & S = \ominus \\ 0 & S = 0, x_2 = \oplus \\ 0 & S = 0, x_2 = \ominus \\ 0 & S = 0 \end{cases}$

$\therefore x(t) = \begin{cases} x_1(t) = 1.5 x_2(t)^2 - 0.5 \\ x_2(t) = -0.5 x_2(t) - 0.5 \end{cases}$

$t_f =$

②

$$H = x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{2}u^2 + \lambda_1 x_2 - \lambda_2 x_1 + \lambda_2 (1-x_1^2)x_2 + \lambda_2 u$$

2H 항은

$$\dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = -(2x_1 - \lambda_2 - 2\lambda_2 x_2 x_1)$$

costate 식.

$$\dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -(x_2 + \lambda_1 + \lambda_2(1-x_1^2))$$

2-1.

u is not bounded, H is quadratic about u .

$$\rightarrow \frac{\partial H}{\partial u} = 0 = u + \lambda_2 \quad \text{즉} \quad \boxed{u = -\lambda_2}$$

2-2.

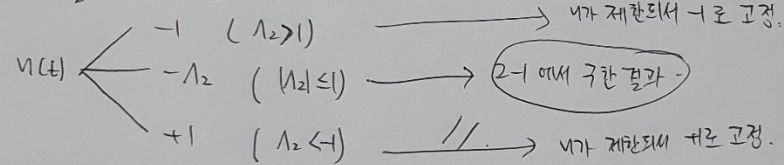
u is bounded, $|u| \leq 1$.

$|\lambda_2| \leq 1$, u 의 최소값은 2-1에 보여줌.

$|\lambda_2| > 1$ 일 때에는 살펴보자.

u 를 선택하되 H 가 음수가 되도록 선택한다. x 와 λ = 고정.

즉, $\frac{1}{2}u^2 + \lambda_2 u$ 을 u 에 대하여 최소화한다.

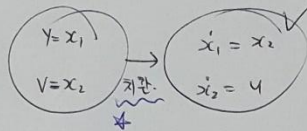


$$\textcircled{3} \Delta T = (1/w_y + T^3/3)(1/w_y + T) - \frac{T^4}{4}$$

$$V(t) = - \left[\frac{(T-t)/w_v + (T-t)^2/2}{\Delta(T-t)} \right] Y(t) - \left[\frac{1/w_y + (T-t)^2/w_v + (T-t)^3/3}{\Delta(T-t)} \right] V(t).$$

$$\dot{Y} = V$$

$$\dot{V} = V$$



$$T = t_f.$$

$$\begin{aligned} w_{x_1} &= w_{x_1} \\ w_v &= w_v \\ w_y &= w_y \end{aligned}$$

$$J = \frac{w_{x_1}}{2} x_1^2(t_f) + \frac{w_{x_2}}{2} x_2^2(t_f) + \frac{1}{2} \int_{t_0}^{t_f} V^2 dt.$$

$$t_f = t_0 + \frac{D}{V}$$

$$\Delta t_f = \left(\frac{1}{w_{x_1}} + \frac{t_f^3}{3} \right) \left(\frac{1}{w_{x_2}} + t_f \right) - \frac{t_f^4}{4}$$

$$V(t) = - \left[\frac{\frac{(t_f-t)}{w_{x_2}} + \frac{(t_f-t)^2}{2}}{\Delta(t_f-t)} \right] x_1(t) - \left[\frac{\frac{1}{w_{x_1}} + \frac{(t_f-t)^2}{w_{x_2}} + \frac{(t_f-t)^3}{3}}{\Delta(t_f-t)} \right] x_2(t)$$

$$f = \dot{x}$$

$$\textcircled{3-1} H = \partial + \lambda f = V^2 + \lambda_1 \dot{x}_1 + \lambda_2 \dot{x}_2 = V^2 + \lambda_1 x_2 + \lambda_2 V = H.$$

$$\dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = 0 \rightarrow \lambda_1 = \text{const.} \quad \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -\lambda_1 \rightarrow \lambda_2 = -\lambda_1 t + \lambda_{20}$$

$$\dot{x} = f = \frac{\partial H}{\partial \lambda} = x_2 + V$$

$$\frac{\partial H}{\partial V} = 0 = \lambda_2 + 2V \quad V = -0.5/\lambda_2$$

3-2

$$\frac{w_{x_1}}{2} x_1^2(t_f) + \frac{w_{x_2}}{2} x_2^2(t_f) \quad h \text{ 라 하면} \quad \frac{\partial h}{\partial x(t_f)} = \lambda^T(t_f).$$

$$\begin{aligned} \frac{\partial h}{\partial x(t_f)} &= \lambda^T(t_f) = w_{x_1} x_1(t_f) \\ \frac{\partial h}{\partial x_2(t_f)} &= \lambda^T(t_f) = w_{x_2} x_2(t_f) \end{aligned} \quad \text{BC 조건.} \quad \begin{aligned} x_1(t_f) &= 0 \\ x_2(t_f) &= 0 \end{aligned}$$

3-3 극값인 $u(t)$ 를 증명.